
✓ Progress Tracker — 10시간 임베디드 제어(HW) 사전학습

Session 1 (2h) — MCU 구조 & 회로 구성 기초

- ☐ Can name the 4 main MCU blocks (CPU, Flash, SRAM, I/O)
- ☐ LED circuit built and blinking, no shorts (Tinkercad + physical)
- ☐ Can explain why the resistor is required

Session 2 (2h) — GPIO

- ☐ Understand INPUT vs INPUT_PULLUP
- ☐ Button toggles LED cleanly, no flicker (Tinkercad + physical)
- ☐ Understand GPIO as the base layer for ADC/interrupts

Session 3 (2h) — ADC/PWM & 센서 연동

- ☐ Understand analogRead() range and analogWrite() PWM pins
- ☐ Sensor/potentiometer smoothly controls LED brightness (Tinkercad + physical)
- ☐ Understand the analogRead → map() → analogWrite() pattern

Session 4 (2h) — 인터럽트

- ☐ Understand why interrupts exist vs. polling
- ☐ Understand `volatile` and why it's required
- ☐ Interrupt-driven button counter works reliably (Tinkercad + physical)

Session 5 (2h) — 통신 & Capstone

- ☐ Understand Serial.begin/print/read
- ☐ Capstone (`integrated_sensor_system.ino`) runs correctly end to end
- ☐ Verified in Tinkercad, then replicated on physical breadboard
- ☐ Can point to where MCU/GPIO/ADC/PWM/interrupt/communication each happen in the code

Ready for 임베디드 제어(HW) internship prerequisite when all boxes above are checked.
